09 - Ohm's Law and Simple Circuit Design

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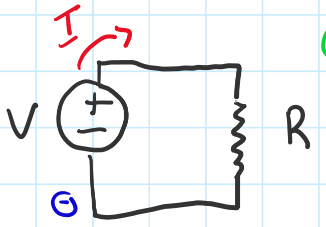
Understanding how Circuits are Designed:

V is like pressure
I is like how much us
R slows it d n

Ex. 1. Voltage applied across resistor



Current flows opposite of e^-

OHM's Law

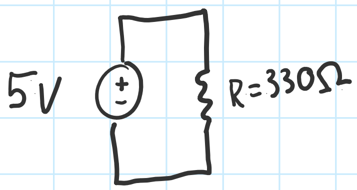
$$\boxed{V = IR}$$

V: Volts, I: Amps, R: Ω OHMs

$I \propto V$

$I \propto \frac{1}{R}$

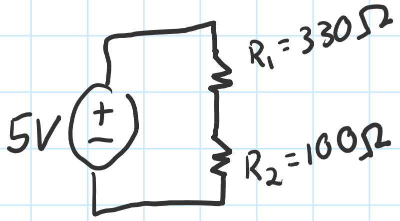
$$\boxed{I = \frac{V}{R}}$$



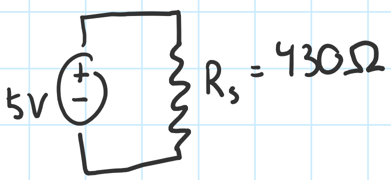
$$I = \frac{V}{R} = \frac{5V}{330\Omega} = \boxed{15.15\,mA}$$

Microprocessors are usually mA or μA

Ex. 2 Resistors in Series



$$R_s = \sum_{i=0}^{n} R_i = 330 + 100 = 430$$



$$I = \frac{V}{R_s} = \frac{5V}{430\Omega} = \boxed{11.63\,mA}$$

Voltage along R_1

$$V_1 = IR_1 = (11.63 \times 10^{-3} A)(330\Omega) = 3.84V$$

$= 5V$

$$V_2 = I R_2 = (11.63 \times 10^{-3} A)(100\,\Omega) = 1.16\,V$$

= 5V